

How to Pick a Model: Information, Allocation, and Proper Time for Verifiable Tasks

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1 Introduction

A builder of agent systems repeatedly faces the same decision: given a verifiable task, a deployment budget, and a menu ranging from inexpensive specialists to frontier generalists, which system should be run? The choice extends beyond the backbone model to the solving strategy, the information available at deployment, and the allocation of compute across repeated attempts. The relevant objective is therefore to identify the configuration that reaches a verified solution most efficiently.

The usual answer is benchmark racing: fix the workflow, compare aggregate scores, and select the winner. This ranks systems but does not explain the ranking. A system may win because each step is cheaper, because it needs fewer steps on the relevant instances, because an observable signal identifies the appropriate strategy, or because its allocation policy wastes less compute. Those mechanisms imply different interventions, yet an aggregate score combines them into one number. Related questions arise in algorithm portfolios, per-instance selection, and model routing [6, 21, 4, 13].

We study this problem for verifiable tasks, including code generation against tests, formal proof search, and verifier-backed planning. Success is determined by an external verifier, allowing performance to be measured by *proper time*: the wall-clock time, tokens, or cost consumed before the first verified solution. This operationalizes the time-to-solution perspective of Achille and Soatto [1] for finite-budget model selection.

We represent task heterogeneity through solver-relevant modes, each associated with a specialized solving strategy. The worker model determines the cost and competence of executing these strategies. Before allocating compute, a router observes a signal Z , such as task metadata, demonstrations, or partial traces. This induces a posterior π_Z over modes, while the router implements an allocation q_Z of the retry budget across strategies. Models and modes therefore remain distinct objects.

Under a packed inverse-share law, the expected log proper-time gain satisfies

$$\Delta_{\log} = I(S; Z) - \mathbb{E}_Z[\text{KL}(\pi_Z \| q_Z)].$$

The first term is the speed potentially available from the signal; the second is the amount lost through allocation mismatch. The identity also prescribes the optimal

allocation under the packed law: $q_Z = \pi_Z$.

When worker models differ, the decomposition becomes

$$\Delta_{M,\log} = \log \frac{\kappa(M_0)}{\kappa(M)} + \varphi(M_0, M) + G(M) - \varepsilon(M),$$

separating per-step cost, mode-conditioned competence, information, and allocation mismatch. It provides the quantities required to decide when a larger model, cheaper retries, or signal-conditioned strategy allocation minimizes verified time.

We evaluate the framework on HumanEval+ using three worker models, three specialized strategies, and twenty attempts per model–strategy–task configuration, for 29,520 verified attempts. After selecting the router and allocation rule on validation data, the resulting policy achieves a 1.113× solver-time speedup on a disjoint test set at fixed 7B worker and unchanged verified success (37/42 tasks under both policies). This comparison isolates allocation speedup and excludes router inference latency.

Contributions.

1. We formulate model selection for verifiable tasks as the joint choice of model, strategy, and retry allocation under a deployment budget.
2. We prove that information-driven expected log-speedup equals signal information minus allocation mismatch under the packed law.
3. We extend the identity to separate model cost, competence, information, and mismatch, obtaining model and retry crossover criteria.
4. We introduce a cross-fitted measurement protocol and evaluate it on HumanEval+ using verified first-passage traces and held-out routing.

2 A Mathematical Framework for Agentic Tasks

2.1 Tasks, Solvers, and Agent Systems

Definition 2.1 (Verifiable task). A *verifiable task* is a tuple

$$\mathcal{T}^{(v)} = (\mathcal{X}, \mathcal{Y}, V, \mu, \mathcal{B}),$$

where $\mathcal{X}$ is the instance space, $\mathcal{Y}$ is the solution space, $V : \mathcal{X} \times \mathcal{Y} \rightarrow \{0, 1\}$ is an external verification oracle that returns 1 exactly when a candidate satisfies the task’s

acceptance criterion, μ is the deployment distribution over $\mathcal{X}$, and $\mathcal{B} = (b_t, b_c, b_m)$ specifies time, cost, and memory budgets. Success on instance x means producing y such that $V(x, y) = 1$ within $\mathcal{B}$.

Definition 2.2 (Agent system and induced solver). For a verifiable task $\mathcal{T}^{(v)}$, an *agent system* is specified by

$$d = (M, \mathcal{H}, \rho) \in \mathcal{D}, \quad (1)$$

where M is the worker model, $\mathcal{H}$ is a finite set of solving strategies or harnesses, and ρ allocates the available attempt budget across those strategies using deployment-side information. These components induce a stochastic solver A_d that executes model-strategy attempts and emits candidate artifacts until V accepts one or the budget is exhausted.

2.2 Proper Time and Certified Speed

Fix a resource clock c : wall-clock time, tokens, API cost, verifier calls, or another declared scalar resource.

Definition 2.3 (Proper time and certified proper time). For solver A on (x, r) , define its *operational proper time* by

$$\tau_A(x; r) := \inf\{c \geq 0 : V(x, Y_A(c; x, r)) = 1\}, \\ \inf \emptyset := \infty.$$

and its *certified proper time* at confidence $1 - \delta$ by

$$\bar{T}_A(\delta) := \inf\{c \geq 0 : \mathbb{P}_{X \sim \mu, r}[\tau_A(X; r) \leq c] \geq 1 - \delta\}.$$

The former is trajectory-level; the latter is its deployment quantile over $X \sim \mu$ and r . Its empirical estimate is the corresponding sample quantile, with failures recorded as ∞ . Relative to baseline A_0 , the *certified log-speedup* is $\Delta := \log \bar{T}_{A_0}(\delta) - \log \bar{T}_A(\delta)$; positive Δ means that A is faster.

The packed-family identities below concern an expected log-time surrogate $\Delta_{\log}$. Certified quantile speedup and operational first-passage speedup are reported separately in the empirical analysis.

2.3 The Packed Latent-Mode Family

Definition 2.4 (Packed allocation policy). Let $\mathcal{S}$ be a finite mode set and let Z take values in a signal space $\mathcal{Z}$. A *packed allocation policy* is a map $\rho : \mathcal{Z} \rightarrow \Delta(\mathcal{S})$, written $z \mapsto q_z$, whose coordinate $q_z(s)$ is the fraction of the attempt budget assigned to the strategy specialized for mode s . The signal Z contains the information available when the allocation is chosen. A no-routing baseline uses a fixed allocation q_0 independent of z .

Assumption 2.5 (Packed latent-mode family). Fix a verifiable task $\mathcal{T}^{(v)}$. Assume:

1. **Latent modes.** There is a finite set of modes $\mathcal{S} = \{1, \dots, m\}$ and a latent variable $S \in \mathcal{S}$ with prior π . A mode is a solver-relevant class of instances, defined independently of which model is deployed. An

observable signal $Z \sim P(Z | S)$ represents admissible pre-deployment evidence or partial execution traces, with posterior $\pi_z(s) := \mathbb{P}[S = s | Z = z]$.

2. **One strategy per mode.** There is a strategy set $\mathcal{H} = \{h_s : s \in \mathcal{S}\}$ in which h_s is specialized for mode s and has packed internal scale $t_0(s) > 0$.
3. **Allocation determines time.** A routing policy chooses a share vector $q \in \Delta(\mathcal{S})$. Under hidden mode s , allocating share $q(s)$ yields certified time

$$\mathsf{T}_q(s) = c_{\text{sim}} \cdot \frac{t_0(s)}{q(s)},$$

where $c_{\text{sim}} > 0$ is the simulation overhead.

4. **Baseline and deployed schedulers.** The baseline chooses a fixed share vector $q_0 \in \Delta(\mathcal{S})$; the deployed system chooses a data-dependent map $z \mapsto q_z \in \Delta(\mathcal{S})$.

All KL and logarithmic expressions below use the standing support convention that every allocation puts positive mass on the posterior support. For a posterior π_z over modes, define $M_{\text{eff}}(z) := \exp(H(\pi_z))$.

Given the feasible designs $d = (M, \mathcal{H}, \rho)$, the builder minimizes certified proper time over model, strategy set, and allocation policy. The following decomposition identifies the mechanisms determining this objective.

3 Information-Routing and Model-Aware Decompositions

3.1 The Routing-Information Identity

Lemma 3.1 (Posterior-matched allocation). *Under Assumption 2.5, for P_Z -almost every z and every $q \in \Delta(\mathcal{S})$ such that $\text{supp}(\pi_z) \subseteq \text{supp}(q)$,*

$$\mathbb{E}_{S \sim \pi_z} \left[\log \frac{\mathsf{T}_q(S)}{c_{\text{sim}} t_0(S)} \right] = H(\pi_z) + \text{KL}(\pi_z \| q).$$

The unique minimizer of $\mathbb{E}_{S \sim \pi_z} [\log \mathsf{T}_q(S)]$ over admissible allocations is $q = \pi_z$. The corresponding excess packed log-time is $\text{KL}(\pi_z \| q)$.

Proof. See Appendix B.1.

Theorem 3.2 (Routing-Information Identity). *Under Assumption 2.5, with a prior-matched baseline $q_0 = \pi$,*

$$\Delta_{\log} = G - \varepsilon$$

where

$$\begin{aligned} \Delta_{\log} &:= \mathbb{E}[\log \mathsf{T}_{\pi}(S) - \log \mathsf{T}_{q_Z}(S)], \\ G &:= \mathbb{I}_{\pi}(S : Z) = H(\pi) - \mathbb{E}_Z[H(\pi_Z)], \quad (2) \\ \varepsilon &:= \mathbb{E}_Z[\text{KL}(\pi_Z \| q_Z)]. \quad (3) \end{aligned}$$

The term G is the mutual information between the latent mode and the feedback—the information generation quality. The term ε is the expected KL divergence of the routing policy from the posterior—the routing mismatch. Equivalently: $\Delta_{\log} \leq G$, with equality if and only

if $q_Z = \pi_Z$ almost surely. One nat of routing mismatch costs exactly one nat of expected packed log-speedup.

Proof. See Appendix B.2.

For a non-prior-matched baseline $q_0 \neq \pi$, the identity becomes $\Delta_{\log} = G + \text{KL}(\pi \| q_0) - \varepsilon$, adding a baseline suboptimality bonus.

3.2 The Model-Aware Decomposition

Assumption 3.3 (Model-dependent packed family). Fix a verifiable task and a shared latent mode variable $S \sim \pi$ over $\mathcal{S} = \{1, \dots, m\}$. For each model M :

1. The per-step cost satisfies $\kappa(M) > 0$.
2. The within-mode packed scale satisfies $t_0(M, s) > 0$, so that packed time under mode s with allocation share $q(s)$ is

$$\mathsf{T}_q^{(M)}(s) = \kappa(M) \cdot c_{\text{sim}} \cdot \frac{t_0(M, s)}{q(s)}.$$

3. The feedback signal is $Z_M \sim P_M(Z | S)$, with posterior $\pi_z^M(s) := \mathbb{P}[S = s | Z_M = z]$ for a realization z .
4. The routing policy is $q_z^M \in \Delta(\mathcal{S})$ for each realized signal z , with $\text{supp}(\pi_z^M) \subseteq \text{supp}(q_z^M)$ almost surely.

Theorem 3.4 (Model-Aware Decomposition). *Under Assumption 3.3, comparing a prior-matched baseline using model M_0 against a deployed system using model M ,*

$$\Delta_{M, \log} = \log \frac{\kappa(M_0)}{\kappa(M)} + \varphi(M_0, M) + G(M) - \varepsilon(M) \quad (4)$$

where the four terms are defined by

$$\Delta_{M, \log} := \mathbb{E} \left[\log \mathsf{T}_\pi^{(M_0)}(S) - \log \mathsf{T}_{q_Z}^{(M)}(S) \right],$$

$$C_\kappa(M_0, M) := \log \frac{\kappa(M_0)}{\kappa(M)}, \quad (5)$$

$$\varphi(M_0, M) := \mathbb{E}_\pi \left[\log \frac{t_0(M_0, S)}{t_0(M, S)} \right], \quad (6)$$

$$G(M) := \mathbb{I}_\pi(S : Z_M), \quad (7)$$

$$\varepsilon(M) := \mathbb{E}_{Z_M} [\text{KL}(\pi_Z^M \| q_Z^M)]. \quad (8)$$

Here C_κ , φ , G , and ε denote per-step cost, within-mode competence, information quality, and routing mismatch, respectively.

Proof. See Appendix B.3.

Appendix Proposition B.1 gives the more general identity in which per-step cost may also vary by mode and the baseline allocation need not equal π .

Benchmark racing reports only an aggregate comparison. The decomposition reports why the expected log-time gain has its value.

3.3 Scope and Metric Guardrails

The preceding identities hold exactly under the packed latent-mode assumptions and concern expected packed

log-time. In particular,

$$\mathbb{E}[\log T_0 - \log T_1] \neq \log \frac{\mathbb{E}[T_0]}{\mathbb{E}[T_1]} \quad \text{in general.}$$

If the packed log-time law has a uniform additive residual $|r_M| \leq \eta$, Proposition B.2 adds a single term R satisfying $|R| \leq 2\eta$.

4 Consequences for Model Choice and System Design

Corollary 4.1 (Model-selection crossover). *Under Assumption 3.3, model M_1 achieves lower expected packed log-time than model M_2 whenever*

$$\log \frac{\kappa(M_2)}{\kappa(M_1)} + \varphi(M_0, M_1) - \varphi(M_0, M_2) > G(M_2) - G(M_1) + \varepsilon(M_1) - \varepsilon(M_2). \quad (9)$$

Proof. See Appendix B.7.

Theorem 4.2 (Single-mode retry crossover). *Fix one homogeneous mode and a common one-attempt horizon. Let a stronger system have success probability p_h and per-attempt cost κ_h , and let a cheaper system have $0 < p_\ell < p_h < 1$ and $0 < \kappa_\ell < \kappa_h$. Assume retries of the cheaper system are conditionally independent and identically distributed. The smallest retry depth matching the stronger system’s one-attempt reliability is*

$$D_{\text{cross}} = \left\lceil \frac{\log(1 - p_h)}{\log(1 - p_\ell)} \right\rceil.$$

Executing all D_{cross} attempts is cheaper if and only if $D_{\text{cross}} \kappa_\ell < \kappa_h$. With stopping at first success, the expected cost is

$$\kappa_\ell \frac{1 - (1 - p_\ell)^{D_{\text{cross}}}}{p_\ell}.$$

Proof. See Appendix C.1.

The inner routing problem of the packed family is the same log-optimal allocation problem that appears in Kelly betting [9], universal portfolio theory [2], and algorithm-portfolio scheduling [6]. The optimal allocation is $q^*(\cdot | Z) = \pi_Z(\cdot)$; the mismatch ε is the expected log-loss from suboptimal allocation.

5 Application and Measurement

5.1 Experimental Design

We evaluate the framework on all 164 HumanEval+ tasks. Before any model run, we partition tasks into three equally sized modes using reference-solution length, with cutoffs at 35 and 58 tokens. The mode labels are used to fit and evaluate posteriors but are not supplied as deployment labels. We pair the modes with three strategy-specific harnesses: a direct implementation prompt, a structured specification-and-invariants prompt, and a robust decomposition and edge-case prompt.

We execute Qwen2.5-Coder 1.5B, 7B, and 32B with each strategy for twenty independent sampling attempts

Table 1: Four-term accounting with twenty in-context examples. Terms are in nats relative to the 1.5B fixed-allocation baseline. “Pred.” is $C_\kappa + \varphi + G - \varepsilon$; “Obs.” is replayed packed log-speedup.

Model	C_κ	φ	G	ε	Pred.	Obs.
1.5B	0.000	0.000	0.128	0.023	0.106	0.128
7B	0.008	0.381	0.128	0.052	0.464	0.319
32B	-1.006	0.879	0.128	0.214	-0.212	0.059

per task. This gives $164 \times 3 \times 3 \times 20 = 29,520$ verifier-scored attempts. Every record contains model, strategy, task mode, generated tokens, wall-clock time, and HumanEval+ verification. Models and modes are therefore distinct: the model controls attempt cost and competence, while the allocation q_Z distributes a twenty-retry budget among strategies.

The router observes task metadata and either 0, 5, or 20 cross-fitted examples. Its calibrated output estimates π_Z ; an allocation head converts this posterior into integer strategy counts summing to twenty. The fixed baseline is $q_0 = (7, 6, 7)/20$. Five-fold predictions estimate the decomposition terms. A separate 81/41/42 train/validation/test split is used once to select the router configuration and evaluate operational allocation speedup.

5.2 Four-Term Diagnosis

The measured per-token costs are 0.01147, 0.01138, and 0.03137 seconds for the 1.5B, 7B, and 32B workers. The inverse-share regression has slope 0.661 in seconds and 0.707 in tokens, compared with the ideal packed value of 1. We therefore report the exact packed accounting together with its observed closure residual rather than identifying the packed law with the operational first-passage process.

For the 7B system, competence is the largest positive term. The signal adds 0.128 nat and the implemented allocation loses 0.052 nat to mismatch. The predicted packed gain is 0.464 nat, whereas the replayed packed gain is 0.319 nat, giving residual -0.145 nat. Operationally, the complete 1.5B-baseline-to-7B-routed comparison gives $\exp(0.400) = 1.49\times$ solver speedup and changes verified success from 86.0% to 93.3%. This system comparison includes the 7B competence gain and is not a router-only effect.

5.3 Held-Out Allocation Test

To isolate routing, we hold the worker model fixed. Router model, context length, and allocation head are chosen on the validation split subject to at most one fewer verified success than the baseline. Only the selected configuration is then evaluated on the 42 untouched test tasks.

At fixed 7B worker, the selected GPT-5.5 router with five in-context examples and a performance-aware allocation head reduces arithmetic mean solver time from 12.00 to 9.54 seconds while preserving all 37 verified successes. The preregistered paired log metric gives the

Table 2: Confirmatory paired test. Speedup is the geometric mean of paired solver-time ratios; intervals are 95% task-bootstrap intervals.

Worker	Speedup	95% CI	Solved
Qwen 1.5B	0.951 $\times$	[0.855, 1.060]	34/42 $\rightarrow$ 34/42
Qwen 7B	1.113 $\times$	[1.007, 1.235]	37/42 $\rightarrow$ 37/42
Qwen 32B	0.855 $\times$	[0.742, 0.961]	36/42 $\rightarrow$ 36/42

1.113 $\times$ speedup in Table 2. Router inference is excluded from this solver-time comparison. The batched router call took 102.5 seconds for 42 tasks, or 2.44 seconds per task under simple amortization; charging that amount eliminates almost all of the arithmetic-mean end-to-end gain.

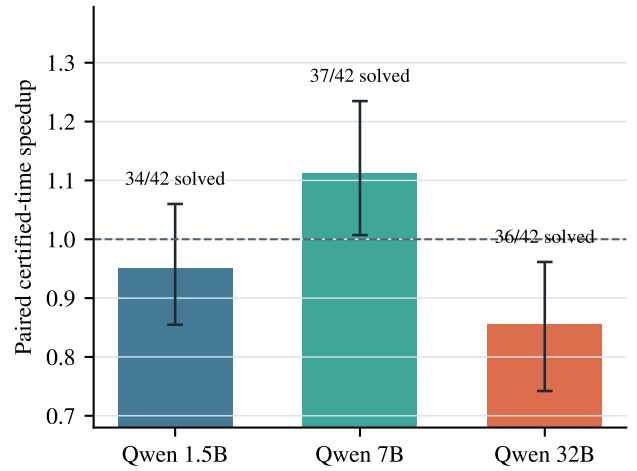


Figure 2: **Held-out allocation speedup.** Bars report paired solver-time speedup after validation-only configuration selection; error bars are 95% task-bootstrap intervals.

5.4 Estimators and Paired Validation

For calibrated posteriors $\hat{\pi}_{Z_i}^M$ and realized allocations $\hat{q}_i^M$, we estimate

$$\hat{G} = H(\hat{\pi}) - \frac{1}{N} \sum_i H(\hat{\pi}_{Z_i}^M), \quad \hat{\varepsilon} = \frac{1}{N} \sum_i \text{KL}(\hat{\pi}_{Z_i}^M \| \hat{q}_i^M),$$

and estimate φ from mode-conditioned packed scales. Temperature calibration and every allocation-head fit use only training or validation records. Task bootstrap preserves the paired comparison between baseline and routed allocations.

Proposition 5.1 (Paired allocation-gain estimator). *Let $X_1, \dots, X_N$ be i.i.d. deployment instances. For condition $k \in \{0, j\}$, let a fixed policy select action $a_i^k = \rho_k(Z_k(X_i))$, where an action may be a retry allocation. Let $\ell(a, x; \xi)$ be an integrable deployment loss, and let $\hat{m}_j(x)$ be an unbiased estimate of the incremental signal cost $m_j(x)$ in the same units as ℓ . If each of Q repeated runs has the corresponding deployment distribution*

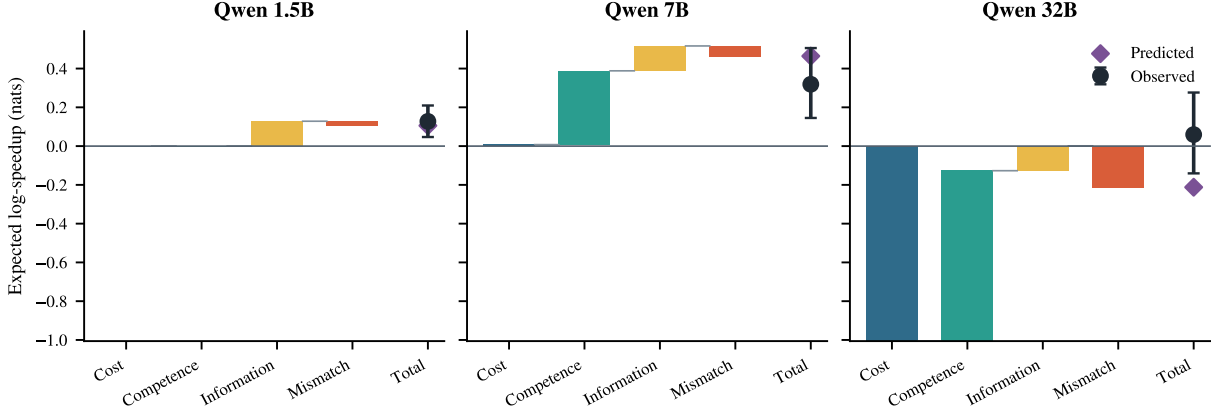


Figure 1: **Diagnosing verified speed.** Bars show per-step cost, competence, information, and mismatch; diamonds show the packed prediction and circles the observed packed log-speedup with task-bootstrap intervals.

conditional on (X_i, a_i^k) , then

$$\bar{\ell}_i^k := \frac{1}{Q} \sum_{q=1}^Q \ell(a_i^k, X_i; \xi_{i,q}^k),$$

$$\hat{\Delta}_R(j) := \frac{1}{N} \sum_{i=1}^N [\bar{\ell}_i^0 - \bar{\ell}_i^j - \hat{m}_j(X_i)]$$

is unbiased for the expected deployment gain net of signal cost.

Proof. See Appendix C.2.

6 Related Work and Positioning

Time and computational information. Achille and Soatto make time-to-solution central for verifiable tasks and connect reusable information to speed [1]. Earlier foundations include universal and resource-bounded search [18, 10, 12, 11, 15]. We adopt the time-centric objective and study a different question: how the deployment-time difference between two agent designs separates into cost, competence, information, and allocation mismatch.

Selection, routing, and test-time compute. Algorithm portfolios and SATzilla select algorithms per instance rather than committing to one global solver [6, 21]. FrugalGPT and RouteLLM bring related cost-quality routing objectives to language models [4, 13]. Adaptive test-time compute further shows that the useful amount of inference effort depends on the prompt and the policy used to spend it [17]. These works motivate the decision variables in our framework. Our policy-agnostic identities expose the terms traded by any allocation policy under the packed-family assumptions.

Sequential and modular computation. The retry analysis is related to anytime algorithms, metareasoning,

and sequential inspection [22, 7, 14, 20, 19, 3]. Mixture-of-experts and conditional computation make routing quality an architectural concern [8, 16, 5]. Our setting differs by allocating an explicit retry budget among complete solving strategies and measuring time to external verification.

7 Discussion and Conclusion

The framework centers on externally verifiable packed latent modes with $\mathbf{T}_q(s) \propto t_0(s)/q(s)$. Uniform η -approximate log-linearity gives residual at most 2η . HumanEval+ uses saved-trace replay; prospective paired deployment additionally charges fresh routing-information acquisition.

Open questions include dynamic mode discovery, correlated strategies, end-to-end router-cost accounting, and direct tests of the inverse-share law beyond the evaluated retry horizon.

The framework gives a decomposable objective, model-selection crossovers, and a measurement protocol, organizing agent-design decisions around named, estimable channels.

A General Agent-System Representation

Definition A.1 (General agent-system representation). For workflows that include tools, persistent memory, context management, and parallel execution, the reduced design tuple of Definition 2.2 may be extended to

$$d_{\text{gen}} = (M, \rho, \Omega, W, K, p),$$

where Ω is the tool library, W is persistent memory, K is the active context budget, and p is the allowed parallelism. An instrumented run can be represented by a resource-indexed artifact process $Y_{A_{d_{\text{gen}}}}(c; x, r)$ and a run graph $G_{A_{d_{\text{gen}}}}(x; r)$ whose nodes are recorded computational steps and whose edges encode observed data and control dependencies.

B Proofs for the Main Text

B.1 Proof of Lemma 3.1: Posterior-matched allocation

Proof. By the inverse-share law, $\log(\mathbb{T}_q(s)/(c_{\text{sim}}t_0(s))) = -\log q(s)$. Hence

$$\mathbb{E}_{S \sim \pi_z} \left[\log \frac{\mathbb{T}_q(S)}{c_{\text{sim}}t_0(S)} \right] = \sum_s \pi_z(s) \log \frac{1}{q(s)} = H(\pi_z) + \text{KL}(\pi_z \| q),$$

where the support condition makes every displayed term finite. The intrinsic term $\mathbb{E}_{S \sim \pi_z} [\log(c_{\text{sim}}t_0(S))]$ is independent of q . Finally, $\text{KL}(\pi_z \| q) \geq 0$, with equality if and only if $q = \pi_z$. This proves both uniqueness and the excess-loss identity. $\square$

B.2 Proof of Theorem 3.2: Routing-Information Identity

Proof. Apply Lemma 3.1 to the baseline allocation $q_0 = \pi$:

$$\mathbb{E}[\log \mathbb{T}_\pi(S)] = \mathbb{E}_\pi[\log(c_{\text{sim}}t_0(S))] + H(\pi).$$

Apply it to the deployed allocation q_Z :

$$\mathbb{E}[\log \mathbb{T}_{q_Z}(S)] = \mathbb{E}_\pi[\log(c_{\text{sim}}t_0(S))] + \mathbb{E}_Z[H(\pi_Z)] + \mathbb{E}_Z[\text{KL}(\pi_Z \| q_Z)].$$

Subtract:

$$\Delta_{\log} = H(\pi) - \mathbb{E}_Z[H(\pi_Z)] - \mathbb{E}_Z[\text{KL}(\pi_Z \| q_Z)] = G - \varepsilon.$$

$\square$

B.3 Proof of Theorem 3.4: Model-Aware Decomposition

Proof. Write out the log-time:

$$\log \mathbb{T}_q^{(M)}(s) = \log \kappa(M) + \log c_{\text{sim}} + \log t_0(M, s) - \log q(s).$$

For the prior-matched baseline with model M_0 :

$$\mathbb{E} \left[\log \mathbb{T}_\pi^{(M_0)}(S) \right] = \log \kappa(M_0) + \log c_{\text{sim}} + \mathbb{E}_\pi[\log t_0(M_0, S)] + H(\pi).$$

For the deployed system with model M :

$$\begin{aligned} \mathbb{E} \left[\log \mathbb{T}_{q_Z}^{(M)}(S) \right] &= \log \kappa(M) + \log c_{\text{sim}} + \mathbb{E}_\pi[\log t_0(M, S)] \\ &\quad + \mathbb{E}_{Z_M}[H(\pi_Z^M)] + \mathbb{E}_{Z_M}[\text{KL}(\pi_Z^M \| q_Z^M)], \end{aligned}$$

where the last two terms arise from the cross-entropy identity exactly as in Theorem 3.2. Subtracting and grouping:

$$\begin{aligned} \Delta_{M, \log} &= \log \frac{\kappa(M_0)}{\kappa(M)} + \mathbb{E}_\pi \left[\log \frac{t_0(M_0, S)}{t_0(M, S)} \right] + (H(\pi) - \mathbb{E}_{Z_M}[H(\pi_Z^M)]) - \mathbb{E}_{Z_M}[\text{KL}(\pi_Z^M \| q_Z^M)] \\ &= \log \frac{\kappa(M_0)}{\kappa(M)} + \varphi(M_0, M) + G(M) - \varepsilon(M). \end{aligned}$$

$\square$

B.4 Mode-dependent cost extension

Proposition B.1 (Mode-dependent cost extension). *Suppose $\kappa(M, s) > 0$ may depend on both model and mode, and*

$$\mathsf{T}_q^{(M)}(s, z) = c_{\text{sim}} \kappa(M, s) \frac{t_0(M, s)}{q_z(s)}.$$

For a baseline (M_0, q_0) and a routed system (M, q_Z) ,

$$\begin{aligned} & \mathbb{E} \left[\log \mathsf{T}_{q_0}^{(M_0)}(S) - \log \mathsf{T}_{q_Z}^{(M)}(S, Z) \right] \\ &= \mathbb{E}_{S \sim \pi} \left[\log \frac{\kappa(M_0, S)}{\kappa(M, S)} \right] + \mathbb{E}_{S \sim \pi} \left[\log \frac{t_0(M_0, S)}{t_0(M, S)} \right] + \mathsf{I}_\pi(S : Z) + \text{KL}(\pi \| q_0) - \mathbb{E}_Z \text{KL}(\pi_Z \| q_Z). \end{aligned}$$

Proof. Apply the conditional cross-entropy identity to the allocation term. The tower property gives $\mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} [\log \kappa(M, S)] = \mathbb{E}_{S \sim \pi} [\log \kappa(M, S)]$ and the analogous equality for t_0 . Subtracting the baseline expression and using $H(\pi) - \mathbb{E}_Z H(\pi_Z) = \mathsf{I}_\pi(S : Z)$ yields the claim. $\square$

B.5 Residual guarantee under approximate packedness

Proposition B.2 (Residual guarantee under approximate packedness). *Suppose the packed log-time law is only approximate, and represent the fixed baseline by a constant signal value z_0 :*

$$\log \mathsf{T}_q^{(M)}(s, z) = \log \kappa(M) + \log c_{\text{sim}} + \log t_0(M, s) - \log q_z(s) + r_M(s, z, q),$$

with $|r_M(s, z, q)| \leq \eta$ uniformly over the evaluated systems, modes, signals, and allocations. For a fixed baseline allocation q_0 and a routed allocation q_Z ,

$$\begin{aligned} & \mathbb{E} \left[\log \mathsf{T}_{q_0}^{(M_0)}(S) - \log \mathsf{T}_{q_Z}^{(M)}(S, Z) \right] \\ &= \log \frac{\kappa(M_0)}{\kappa(M)} + \varphi(M_0, M) + \mathsf{I}_\pi(S : Z) + \text{KL}(\pi \| q_0) - \mathbb{E}_Z \text{KL}(\pi_Z^M \| q_Z^M) + R, \end{aligned}$$

where

$$R := \mathbb{E}[r_{M_0}(S, z_0, q_0) - r_M(S, Z, q_Z)], \quad |R| \leq 2\eta.$$

For $q_0 = \pi$, this is Theorem 3.4 plus a residual bounded by 2η .

Proof. Substitute the approximate log-time law into the proof of Proposition B.1, specialized to mode-independent $\kappa(M)$. The ideal terms decompose exactly as before. The remaining difference is

$$R = \mathbb{E}[r_{M_0}(S, z_0, q_0) - r_M(S, Z, q_Z)].$$

Therefore

$$|R| \leq \mathbb{E}|r_{M_0}(S, z_0, q_0)| + \mathbb{E}|r_M(S, Z, q_Z)| \leq 2\eta.$$

$\square$

B.6 Cost of a Borrowed Allocation

Proposition B.3 (Cost of a borrowed allocation). *Under Assumption 3.3, fix a model M_B and let $q_Z^{*,B} = \pi_Z^B$. For any allocation policy q_Z satisfying the standing support condition,*

$$\mathbb{E} \left[\log \mathsf{T}_{q_Z}^{(M_B)}(S) - \log \mathsf{T}_{q_Z^{*,B}}^{(M_B)}(S) \right] = \mathbb{E}_Z [\text{KL}(\pi_Z^B \| q_Z)].$$

In particular, if $q_Z = q_Z^{,A}$ and $q_Z^{*,A} \neq q_Z^{*,B}$ on a set of positive probability, then the excess is strictly positive whenever the displayed expectation is finite.*

Proof. Condition on $Z = z$. The model-dependent cost and competence terms are fixed, so Lemma 3.1 gives an excess conditional packed log-time of $\text{KL}(\pi_z^B \| q_z)$. Taking expectation over Z proves the identity. Nonnegativity of KL divergence, with equality only when its arguments agree, gives the final claim. $\square$

B.7 Proof of Corollary 4.1: Model-selection crossover

Proof. Apply Theorem 3.4 to both models with the same baseline; subtract. The baseline terms cancel. $\square$

C Retry and Paired-Validation Results

Proposition C.1 (Geometric-retry certified scale). *Suppose independent attempts of system M on mode s succeed with probability $p_M(s) \in (0, 1)$ and each complete attempt has duration $u_M(s) > 0$. The minimum number of attempts needed to achieve failure probability at most $\delta \in (0, 1)$ is*

$$N_M(s, \delta) = \left\lceil \frac{\log \delta}{\log(1 - p_M(s))} \right\rceil,$$

and the corresponding focused certified scale is $t_0(M, s, \delta) = u_M(s)N_M(s, \delta)$. Equivalently, the exact success hazard is $h_M(s) := -\log(1 - p_M(s))$; replacing it by $p_M(s)$ is only a small- $p_M(s)$ approximation.

Proof. After N independent attempts, failure has probability $(1 - p_M(s))^N$. Thus $(1 - p_M(s))^N \leq \delta$ if and only if $N \geq \log \delta / \log(1 - p_M(s))$; taking the least integer proves the claim. $\square$

C.1 Proof of Theorem 4.2: Single-mode retry crossover

Proof. The cheaper system succeeds at least once in D attempts with probability $1 - (1 - p_\ell)^D$. Requiring this quantity to be at least p_h and dividing by the negative number $\log(1 - p_\ell)$ gives $D \geq \log(1 - p_h) / \log(1 - p_\ell)$. The ceiling is therefore the least feasible integer depth. The fixed-depth cost statement is immediate. For early stopping, attempt d is reached with probability $(1 - p_\ell)^{d-1}$; summing its expected cost gives the finite geometric series. $\square$

Definition C.2 (Finite-horizon first-hit diagnostics). Fix a horizon H , action a , mode s , and censoring floor $p_{\min} \in (0, 1)$. Let τ_a be first verified success and $C_t(a)$ accumulated deployment cost through time t . Define

$$F_a(s, H) := \mathbb{P}(\tau_a \leq H \mid S = s), \quad C_a(s, H) := \mathbb{E}[C_{\tau_a \wedge H}(a) \mid S = s],$$

Assume $C_a(s, H) > 0$ and define the finite-horizon log-effort diagnostic

$$\ell_{\text{hit}}(a, s; H) := \log C_a(s, H) - \log(F_a(s, H) \vee p_{\min}).$$

This empirical surrogate is neither the certified-time quantile $\bar{T}_A(\delta)$ nor the exact packed time T_q .

C.2 Proof of Proposition 5.1: Paired allocation-gain estimator

Proof. Condition on each sampled instance and its two selected actions. Each repeated-run average has the corresponding conditional expected deployment loss, while $\hat{m}_j$ has conditional expectation m_j . The tower property and linearity of expectation then give the stated equality. $\square$

D Additional Strategy-Routing Diagnostics

All quantities in this section are computed from the saved 29,520 worker attempts. Counterfactual allocations replay the same model–strategy traces under a common twenty-retry horizon; they are not additional model calls.

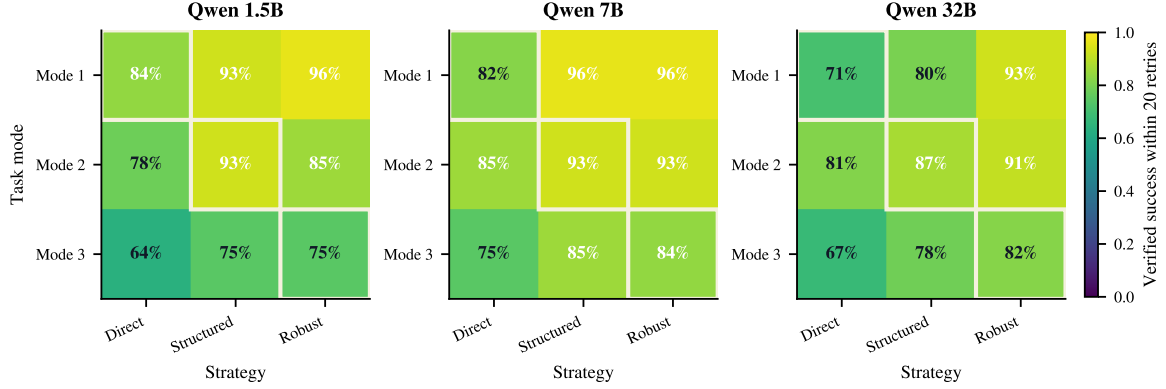


Figure 3: **Mode-conditioned strategy specialization.** Cells report verified success within twenty attempts. Strategy performance varies with task mode and worker model, establishing the heterogeneity required for nontrivial allocation.

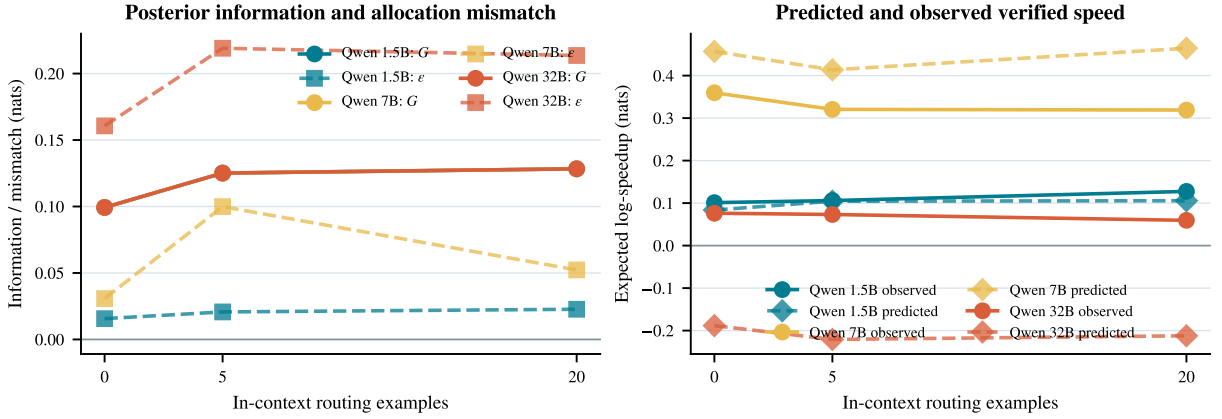


Figure 4: **Information, mismatch, and packed speed.** The left panel reports calibrated information and allocation mismatch as the number of in-context routing examples changes. The right panel compares the resulting packed prediction with replayed packed log-speedup.

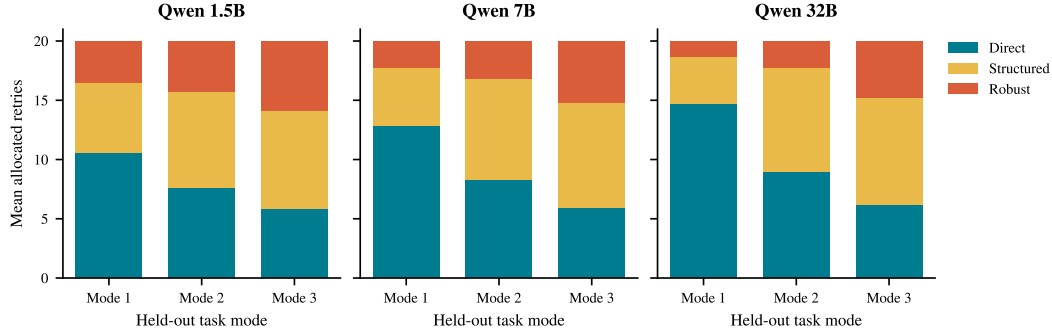


Figure 5: **Mean strategy allocation by held-out task mode.** Every bar contains exactly twenty retries. The allocation changes with both task mode and worker model, while models and modes remain separate experimental objects.

E Theorem-to-Evidence Audit

Exact identities are established by proof. Experiments estimate their terms and test their structural assumptions; they do not replace those proofs.

Result	Status	Evidence and limitation
Posterior-matched allocation and Routing-Information Identity	Partial	Calibrated posteriors and realized strategy shares identify $G = 0.128$ nat and $\varepsilon = 0.052$ nat for the reported 7B configuration. The measured inverse-share slopes, 0.661 in seconds and 0.707 in tokens, differ from the ideal value 1.
Model-Aware Decomposition	Partial	The 1.5B-to-7B accounting estimates cost 0.008, competence 0.381, information 0.128, and mismatch 0.052 nat. Predicted and observed packed gains differ by -0.145 nat.
Model-selection crossover	Partial	Mode-strategy cells have different cost-success profiles. The experiment does not statistically identify a globally optimal worker for every mode.
Single-mode retry crossover	Partial	Twenty repeated attempts estimate retry curves, but conditional i.i.d. retries within each mode have not been established.
Paired allocation-gain estimator	Protocol-aligned	Router configuration is selected on validation data and evaluated once on 42 held-out tasks. The solver-time comparison excludes router inference cost; the measured batched cost is reported separately.
Approximate-packedness residual	Partial	The observed closure residual is finite, but a uniform bound $ r_M \leq \eta$ over all admissible allocations is not established.

F Notation and Assumption Audit

Symbol	Definition
$\mathcal{T}^{(v)} = (\mathcal{X}, \mathcal{Y}, V, \mu, \mathcal{B})$	Verifiable task
$\bar{T}_A(\delta)$	Certified time-to-solution at confidence $1 - \delta$
Δ	Certified log-speedup: $\log \bar{T}_{A_0} - \log \bar{T}_A$
$\mathsf{T}_q^{(M)}(s)$	Packed mode-conditioned time-sharing surrogate; distinct from $\bar{T}_A(\delta)$
$\Delta_{\log}, \Delta_{M,\log}$	Expected packed log-speedup functionals
$S \in \mathcal{S}$	Latent mode variable
π, π_Z	Prior and posterior over modes
q, q_Z	Baseline and deployed routing allocations
$\kappa(M)$	Per-step cost of model M
$t_0(M, s)$	Within-mode packed scale for model M on mode s
$\varphi(M_0, M)$	$\mathbb{E}_\pi[\log(t_0(M_0, S)/t_0(M, S))]$: competence advantage
$G(M)$	$\mathbb{I}_\pi(S : Z_M)$: information generation quality
$\varepsilon(M)$	$\mathbb{E}_Z[\text{KL}(\pi_Z^M \ q_Z^M)]$: routing mismatch
M_{eff}	$\exp(H(\pi_Z))$: entropy-effective mode count
$d = (M, \mathcal{H}, \rho)$	Worker model, strategy set, and allocation policy
$d_{\text{gen}} = (M, \rho, \Omega, W, K, p)$	Extended agent-system coordinates (Appendix A)
r_M	Additive residual from the packed log-time law

Result	Assumptions Required
RI Identity (Thm. 3.2)	Packed family (Asm. 2.5), prior-matched baseline, expected-log functional
Four-term decomposition (Thm. 3.4)	Model-dependent packed family (Asm. 3.3), expected-log functional
Residual guarantee (Prop. B.2)	Uniform η -approximate packed log-linearity; residual at most 2η
Mode-dependent cost extension (Prop. B.1)	Positive mode-dependent costs, packed inverse-share law, common mode prior
Crossover under retries (Thm. 4.2)	Homogeneous mode, i.i.d. retries, common horizon, fixed per-attempt costs
Paired router estimator (Prop. 5.1)	I.i.d. instances, fixed routers, integrable loss, unbiased measurement-cost estimate
Crossover formula (Cor. 4.1)	Model-dependent packed family

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